

Paper 2.pdf
PDF

Paper 1.pdf
PDF

Hi. What is the connection between the definition of RBE as given in these two papers? I would like to derive the expression in paper 2 in such a way that it matches the expression as given in paper 1

Good question—these two papers are actually using the *same underlying definition of RBE*, but expressed at different levels:

- **Paper 2 (Kase et al.)** defines RBE via *dose for equal survival* and plugs in a **specific survival model (MKM)**.
- **Paper 1 (Rørvik et al.)** rewrites the same idea in a **general LQ (linear–quadratic) framework**, making the dependence on parameters explicit.

Let's connect them step by step.

1. Start from the common definition of RBE

Both papers ultimately use:

$$\text{RBE} = \frac{D_{\text{reference}}}{D_{\text{proton}}}$$

for the **same biological endpoint (same survival fraction S)**.

This is explicitly written in Paper 2:

$$\text{RBE} = \frac{D_{S,R}}{D_S}$$

 Paper 2

So the whole problem reduces to:

Express D_S and $D_{S,R}$ using survival models, then take the ratio.

2. Survival model in both papers (this is the key bridge)

Both papers assume the **linear-quadratic (LQ) model**:

$$S = \exp(-\alpha D - \beta D^2)$$

- In **Paper 1**, this is explicit and general  Paper 1
- In **Paper 2**, it's embedded inside the **MKM**, but the final form is still LQ-like:

$$S = \exp(-\alpha D - \beta D^2)$$

where:

- α is modified via microdosimetry (depends on y^*)
- β is taken constant  Paper 2

👉 So **MKM** → **effective LQ parameters**.

This is the crucial connection:

Paper 2 just provides a physics-based way to compute α , while Paper 1 treats α, β as inputs.

3. Solve for dose at a given survival

From the LQ model:

$$S = \exp(-\alpha D - \beta D^2)$$

Take log:

$$\ln S = -\alpha D - \beta D^2$$

Rearrange:

$$\beta D^2 + \alpha D + \ln S = 0$$

Solve quadratic:

$$D = \frac{-\alpha + \sqrt{\alpha^2 - 4\beta \ln S}}{2\beta}$$

4. Apply this to both radiations

Proton beam:

$$D_p = \frac{-\alpha + \sqrt{\alpha^2 - 4\beta \ln S}}{2\beta}$$

Reference radiation:

$$D_x = \frac{-\alpha_x + \sqrt{\alpha_x^2 - 4\beta_x \ln S}}{2\beta_x}$$

5. Form the RBE ratio

$$\text{RBE} = \frac{D_x}{D_p}$$

This gives exactly the structure seen in **Paper 2 Eq. (3)**  Paper 2

6. Now match Paper 1's expression

Paper 1 rewrites the same ratio into a more compact function:

$$\text{RBE}(D_p, \alpha, \alpha_x, \beta, \beta_x)$$

 Paper 1

They eliminate S and express everything in terms of **dose** D_p instead.

How?

From proton survival:

$$\ln S = -\alpha D_p - \beta D_p^2$$

Plug this into the **reference dose expression**:

$$D_x = \frac{-\alpha_x + \sqrt{\alpha_x^2 + 4\beta_x(\alpha D_p + \beta D_p^2)}}{2\beta_x}$$

Then:

$$\text{RBE} = \frac{D_x}{D_p}$$

This algebra leads directly to Paper 1 Eq. (1)  Paper 1

7. Final conceptual connection

Paper 2:

- Starts from survival S
- Uses MKM $\rightarrow$ gives $\alpha(y^*)$
- Computes RBE via **dose-for-fixed-survival**

Paper 1:

- Starts from LQ parameters
 - Eliminates S
 - Expresses RBE directly as a function of **dose** D_p
-

8. Key insight (this is the “bridge” you want)

You can derive Paper 1 from Paper 2 by:

1. Start with:

$$\text{RBE} = \frac{D_x(S)}{D_p(S)}$$

2. Use LQ for both
3. Replace:

$$\ln S = -\alpha D_p - \beta D_p^2$$

4. Substitute into D_x
 5. Divide by D_p
-

9. What differs physically (not mathematically)

- **Paper 1:** phenomenological (fits α, β)
- **Paper 2:** mechanistic (derives α from microdosimetry via y^*)

But mathematically:

They are identical once expressed in LQ form.

If you want, ↪ I can do a clean line-by-line derivation from Eq. (3) in Paper 2 to Eq. (1) in Paper 1 with all algebra shown explicitly.

     ...  Sources

Show me a clean line-by-line derivation from Eq. (3) in Paper 2 to Eq. (1) in Paper 1 with all algebra shown explicitly

Great—let's do this cleanly and explicitly. I'll start from **Eq. (3) in Paper 2** and transform it step by step until it matches **Eq. (1) in Paper 1**.

0. What we are connecting

Paper 2 (Eq. 3):

$$\text{RBE} = \frac{D_{S,R}}{D_S}$$

 Paper 2

Paper 1 (Eq. 1):

$$\text{RBE}(D_p, \alpha, \alpha_x, \beta, \beta_x) = \frac{1}{2D_p} \left[-\frac{\alpha_x}{\beta_x} + \sqrt{\left(\frac{\alpha_x}{\beta_x}\right)^2 + \frac{4D_p}{\beta_x}\alpha + \frac{4D_p^2}{\beta_x}\beta} \right]$$

 Paper 1

Goal: derive this expression starting from the definition.

1. Write survival for both radiations (LQ model)

For protons:

$$S = \exp(-\alpha D_p - \beta D_p^2)$$

For reference photons:

$$S = \exp(-\alpha_x D_x - \beta_x D_x^2)$$

Same S by definition of RBE.

2. Take logarithms

$$\ln S = -\alpha D_p - \beta D_p^2$$

$$\ln S = -\alpha_x D_x - \beta_x D_x^2$$

3. Solve for D_x (reference dose)

Rearrange photon equation:

$$\beta_x D_x^2 + \alpha_x D_x + \ln S = 0$$

Quadratic solution:

$$D_x = \frac{-\alpha_x + \sqrt{\alpha_x^2 - 4\beta_x \ln S}}{2\beta_x}$$

4. Substitute $\ln S$ from proton expression

From step 2:

$$\ln S = -\alpha D_p - \beta D_p^2$$

Insert into D_x :

$$D_x = \frac{-\alpha_x + \sqrt{\alpha_x^2 + 4\beta_x(\alpha D_p + \beta D_p^2)}}{2\beta_x}$$

5. Expand inside the square root

$$D_x = \frac{-\alpha_x + \sqrt{\alpha_x^2 + 4\beta_x\alpha D_p + 4\beta_x\beta D_p^2}}{2\beta_x}$$

6. Form the RBE ratio

From definition:

$$\text{RBE} = \frac{D_x}{D_p}$$

Substitute D_x :

$$\text{RBE} = \frac{1}{D_p} \cdot \frac{-\alpha_x + \sqrt{\alpha_x^2 + 4\beta_x\alpha D_p + 4\beta_x\beta D_p^2}}{2\beta_x}$$

7. Rearrange prefactors

$$\text{RBE} = \frac{1}{2\beta_x D_p} \left[-\alpha_x + \sqrt{\alpha_x^2 + 4\beta_x\alpha D_p + 4\beta_x\beta D_p^2} \right]$$

8. Factor out $1/\beta_x$ inside the bracket

Write everything in terms of α_x/β_x :

$$= \frac{1}{2D_p} \left[-\frac{\alpha_x}{\beta_x} + \sqrt{\frac{\alpha_x^2}{\beta_x^2} + \frac{4\alpha D_p}{\beta_x} + \frac{4\beta D_p^2}{\beta_x}} \right]$$

9. Final result

$$\boxed{\text{RBE} = \frac{1}{2D_p} \left[-\frac{\alpha_x}{\beta_x} + \sqrt{\left(\frac{\alpha_x}{\beta_x}\right)^2 + \frac{4D_p}{\beta_x}\alpha + \frac{4D_p^2}{\beta_x}\beta} \right]}$$

10. Match with Paper 1

This is exactly Eq. (1) in Paper 1, just written in the same structure.  Paper 1

11. Key takeaway

- Paper 2: defines RBE via equal survival + solves for doses
- Paper 1: eliminates S and expresses everything in terms of D_p

👉 The derivation above is precisely that elimination step.

If you want, I can also $\hookrightarrow$ show how this reduces to $\text{RBE}_{\max} = \alpha/\alpha_x$ and $\text{RBE}_{\min} = \sqrt{\beta/\beta_x}$ in the limits $D_p \rightarrow 0$ and $D_p \rightarrow \infty$, which is the next step Paper 1 takes.

     ...  Sources